Solution Set

Question 1

A default indicator D has probability p=0.03. State its distribution, mean, variance, skew intuition, and finance interpretation.

Answer key

D is Bernoulli:

$$D \sim \text{Bernoulli}(0.03)$$

$$\mathbb{E}[D] = p = 0.03, \qquad \text{Var}(D) = p(1-p) = 0.0291$$

The distribution is highly right-skewed because most observations are zero and rare observations are one. It models default/no-default over a fixed horizon.

Question 2

A portfolio has 100 independent names with default probability 2 percent. Use the binomial model to compute mean, variance, and P(no default).

Answer key

The default count is

$$N \sim \text{Binomial}(100, 0.02)$$

$$\mathbb{E}[N] = 100(0.02) = 2$$

$$\text{Var}(N) = 100(0.02)(0.98) = 1.96$$

$$\mathbb{P}(N=0) = 0.98^{100} \approx 0.1326$$

Question 3

Use a Poisson approximation for the default count in Question 2. Compare P(no default).

Answer key

For large n and small p with lambda=np=2,

$$N \approx \text{Poisson}(2)$$

The no-default probability is

$$\mathbb{P}(N=0) = e^{-2} \approx 0.1353$$

This is close to the exact binomial value 0.1326. The approximation is useful because default probabilities are small and name count is large.

Question 4

A continuously compounded return R is normal with mean -1 percent and volatility 3 percent. If S0=100, describe the distribution of S1 and compute its median.

Answer key

If

$$R \sim N(-0.01, 0.03^2), \qquad S_1 = 100e^R$$

then S1 is lognormal. The median is obtained by exponentiating the mean log return:

$$\text{median}(S_1) = 100e^{-0.01} \approx 99.005$$

The mean would be higher than the median because the lognormal distribution is right-skewed.

Question 5

For

$$X \sim N(\mu, \sigma^2)$$

derive parametric VaR for loss

$$L = -X$$

at confidence

$$\alpha$$

Answer key

If X is PnL, loss is L=-X, so

$$L \sim N(-\mu, \sigma^2)$$

The alpha-quantile of L is

$$\text{VaR}_\alpha(L) = -\mu + \sigma z_\alpha$$

where z_alpha is the alpha standard-normal quantile. This formula is simple but fragile under skewness and heavy tails.

Question 6

Derive the exponential survival function from a constant hazard rate lambda.

Answer key

A constant hazard means

$$\lambda = -\tfrac{d}{dt}\log S(t)$$

Integrating from 0 to t gives

$$\log S(t) - \log S(0) = -\lambda t$$

With S(0)=1,

$$S(t) = e^{-\lambda t}$$

This is the default-time model behind a flat hazard-rate credit curve.

Question 7

A liquidity slippage X is uniform on [2,10] bps. Compute mean, variance, and 95 percent quantile.

Answer key

For X ~ U(a,b),

$$\mathbb{E}[X] = \tfrac{a+b}{2} = 6$$

$$\text{Var}(X) = \tfrac{(b-a)^2}{12} = \tfrac{64}{12} = 5.333$$

$$q_{0.95} = a + 0.95(b-a) = 2 + 0.95(8) = 9.6$$

Question 8

Why is the Student t distribution often used as an alternative to normal returns?

Answer key

The Student t distribution has heavier tails than the normal distribution. This assigns more probability to large moves while preserving a symmetric bell shape. The degrees of freedom parameter controls tail thickness; lower degrees of freedom mean fatter tails.

Question 9

For a chi-square variable with n degrees of freedom, state its connection to sample variance.

Answer key

If X_1,...,X_n are iid normal with variance sigma^2 and sample variance S^2, then

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$$

This connects estimation error in volatility to the chi-square distribution.

Question 10

Explain when a Poisson process is a poor model for defaults.

Answer key

A simple Poisson process assumes independent arrivals and constant intensity. It is poor when defaults cluster, intensities change with macro conditions, contagion exists, or firms share latent credit factors. In those cases a Cox process or factor model may be more realistic.

Question 11

A beta distribution is sometimes used for recovery rates. Why is support important here?

Answer key

Recovery rates are naturally bounded between 0 and 1. A beta distribution has support on [0,1], so it respects the variable's domain. A normal model can assign impossible negative recoveries or recoveries above 100 percent unless truncated.

Question 12

Give a model-risk warning for fitting named distributions to market returns.

Answer key

A named distribution is an approximation, not a fact. A good fit in the center can still fail in the tails. Interview-quality validation checks QQ plots, tail exceedances, stability over time, stress periods, and whether the distribution is being used for pricing or real-world risk.